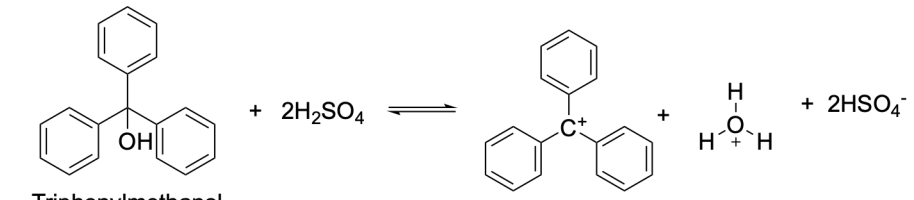
Experiment: Synthesis of Triphenylmethyl Methyl Ether (Trityl Methyl Ether)

Objective:
To synthesize triphenylmethyl methyl ether from triphenylmethanol via formation of the trityl carbocation in concentrated sulfuric acid, isolate the product, determine the percent yield, and evaluate its purity by melting-point analysis before and after recrystallization.

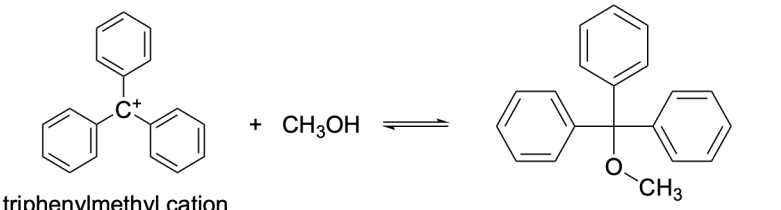
Background:
Triphenylmethanol reacts with concentrated sulfuric acid to form a resonance-stabilized trityl carbocation. The hydroxyl group is first protonated and then eliminated as water, producing the intensely colored carbocation intermediate. Methanol acts as a nucleophile and attacks the carbocation, forming triphenylmethyl methyl ether after deprotonation. The crude product is purified by recrystallization from methanol.



Triphenylmethanol
MW 260.34, mp 163°C

triphenylmethyl cation
orange-yellow

After dissolution in concentrated sulfuric acid, the hydroxyl anion is protonated; then the portion is lost as (which is then itself protonated), leaving the carbocation. The bisulfate ion is a very weak nucleophile and does not compete with methanol in the formation of the product, trityl methyl ether.



triphenylmethyl cation
orange-yellow

triphenylmethyl methyl ether
MW 274.37, mp 96°C

Materials:

- Triphenylmethanol (50 mg)
- Concentrated sulfuric acid (0.4–0.5 mL)
- Ice-cold methanol (3 mL)
- Additional cold methanol for rinsing
- Distilled water

Equipment:

- 10 × 100 mm reaction tubes (2)
- Glass stirring rod
- Pasteur pipette
- Hirsch funnel with polyethylene frit
- Filter paper (for drying only)
- Balance
- Melting-point apparatus

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Safety Precautions:

1. Concentrated sulfuric acid is highly corrosive and can cause severe burns. Handle with care and wear appropriate personal protective equipment.
2. Methanol is toxic and flammable. Avoid inhalation, ingestion, and contact with skin.
3. Always add acid-containing solutions carefully to methanol as directed.
4. Perform the experiment in a well-ventilated area or fume hood.
5. Dispose of chemical waste according to laboratory procedures.

Procedure:

(Reference: eTextbook: Williamson, K.L. and Kenneth L., Macroscale and Microscale Organic Experiments, 7th Edition, Cengage Learning, p. 449)

Part A. Synthesis of Crude Product

1. Weigh out 50 mg of triphenylmethanol and transfer it to a 10 × 100 mm reaction tube.
2. Grind the crystals into a fine powder using a glass stirring rod.
3. Add 0.4–0.5 mL of concentrated sulfuric acid and continue stirring until all of the solid dissolves.
4. Observe and record the color of the solution. Explain the observed color based on the reaction mechanism.
5. Place 3 mL of ice-cold methanol in a second reaction tube.
6. Using a Pasteur pipette, transfer the sulfuric acid solution to the cold methanol.
7. Rinse the first reaction tube with a small amount of cold methanol and add the rinse to the reaction mixture.

Part B. Crystallization and Isolation

1. Allow crystals to form. If crystallization does not occur readily, induce crystallization by one or more of the following methods:
    - Cooling the solution further
    - Adding a seed crystal
    - Scratching the inside wall of the tube at the liquid-air interface with a glass stirring rod
2. Collect the crystals by vacuum filtration using a Hirsch funnel with a polyethylene frit.
3. Do not place filter paper on top of the frit, as the strongly acidic solution may damage the paper.
4. Wash the crystals thoroughly with water.
5. Dry the crystals by pressing them between sheets of filter paper.

Part C. Yield Determination

1. Weigh the crude product.
2. Calculate the theoretical yield and the percent yield.
3. Reserve a small sample for melting-point determination.
4. Measure and record the melting point of the crude product.

Part D. Recrystallization

1. Recrystallize the remaining product from boiling methanol.
2. Collect and dry the purified crystals.
3. Determine and record the melting point of the purified triphenylmethyl methyl ether.

Data and Observations:

Record the following:

- Mass of triphenylmethanol used:
- Color observed after dissolution in sulfuric acid:
- Mass of crude product obtained:
- Theoretical yield:
- Percent yield:
- Melting point of crude product:
- Melting point of purified product:

Cleaning Up:

Dilute the filtrate with water, neutralize with sodium carbonate, and flush down the drain.

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Submit your lab report in the prescribed template. Complete the assignments given on the worksheet and attach it with your lab report.